

```
In [2]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [3]: data = sns.load_dataset('titanic')
```

```
In [4]: print(data.head())
```

|   | survived | pclass | sex    | age  | sibsp | parch | fare    | embarked | class | \ |
|---|----------|--------|--------|------|-------|-------|---------|----------|-------|---|
| 0 | 0        | 3      | male   | 22.0 | 1     | 0     | 7.2500  | S        | Third |   |
| 1 | 1        | 1      | female | 38.0 | 1     | 0     | 71.2833 | C        | First |   |
| 2 | 1        | 3      | female | 26.0 | 0     | 0     | 7.9250  | S        | Third |   |
| 3 | 1        | 1      | female | 35.0 | 1     | 0     | 53.1000 | S        | First |   |
| 4 | 0        | 3      | male   | 35.0 | 0     | 0     | 8.0500  | S        | Third |   |

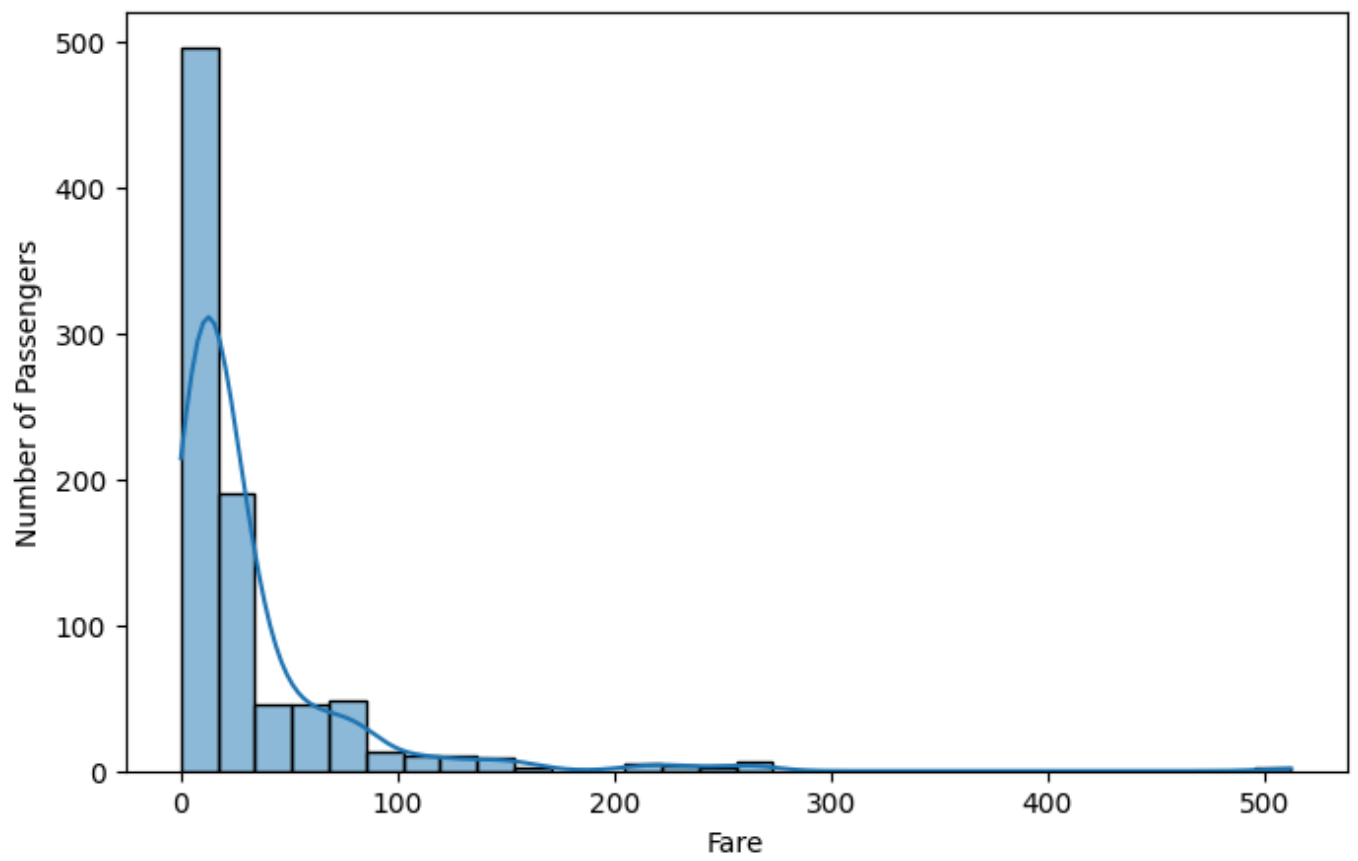
|   | who   | adult_male | deck | embark_town | alive | alone |
|---|-------|------------|------|-------------|-------|-------|
| 0 | man   | True       | NaN  | Southampton | no    | False |
| 1 | woman | False      | C    | Cherbourg   | yes   | False |
| 2 | woman | False      | NaN  | Southampton | yes   | True  |
| 3 | woman | False      | C    | Southampton | yes   | False |
| 4 | man   | True       | NaN  | Southampton | no    | True  |

```
In [5]: print(data.info())
```

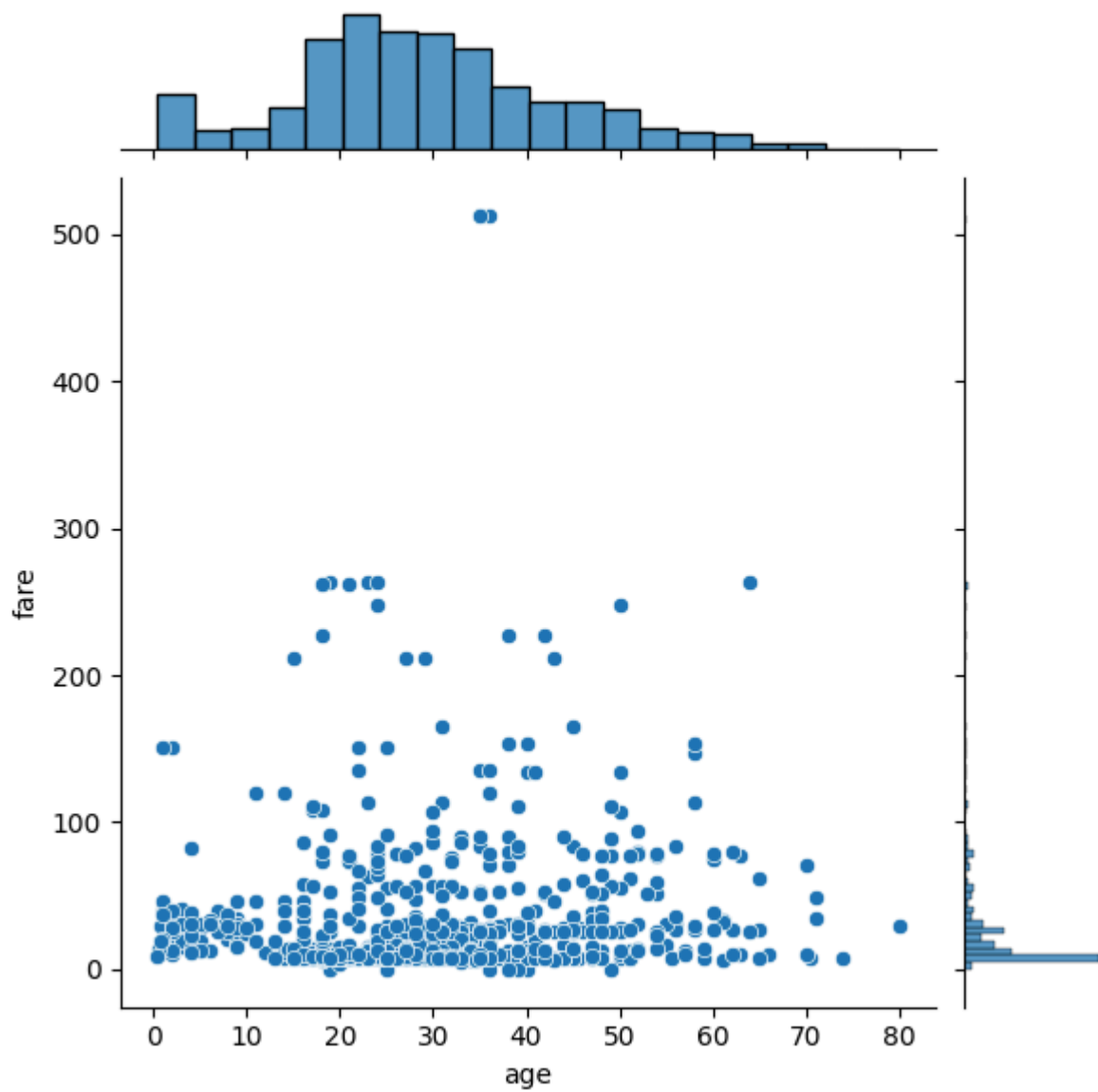
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
#   Column          Non-Null Count  Dtype
---  -
0   survived        891 non-null   int64
1   pclass          891 non-null   int64
2   sex             891 non-null   object
3   age            714 non-null   float64
4   sibsp          891 non-null   int64
5   parch          891 non-null   int64
6   fare           891 non-null   float64
7   embarked       889 non-null   object
8   class          891 non-null   category
9   who            891 non-null   object
10  adult_male     891 non-null   bool
11  deck          203 non-null   category
12  embark_town    889 non-null   object
13  alive         891 non-null   object
14  alone         891 non-null   bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
None
```

```
In [13]: plt.figure(figsize=(8,5))
sns.histplot(data['fare'].dropna(), bins=30, kde=True)
plt.title("Distribution of Ticket Fare")
plt.xlabel("Fare")
plt.ylabel("Number of Passengers")
plt.show()
```

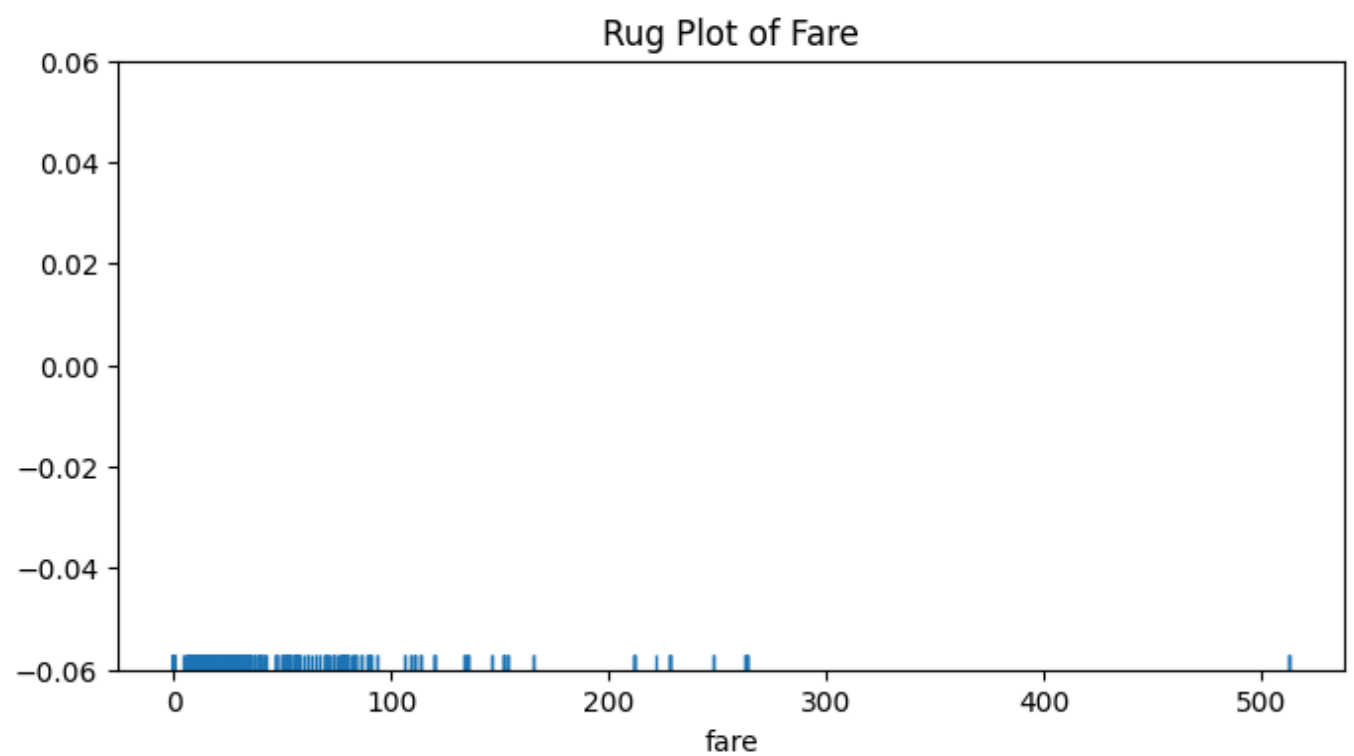
Distribution of Ticket Fare



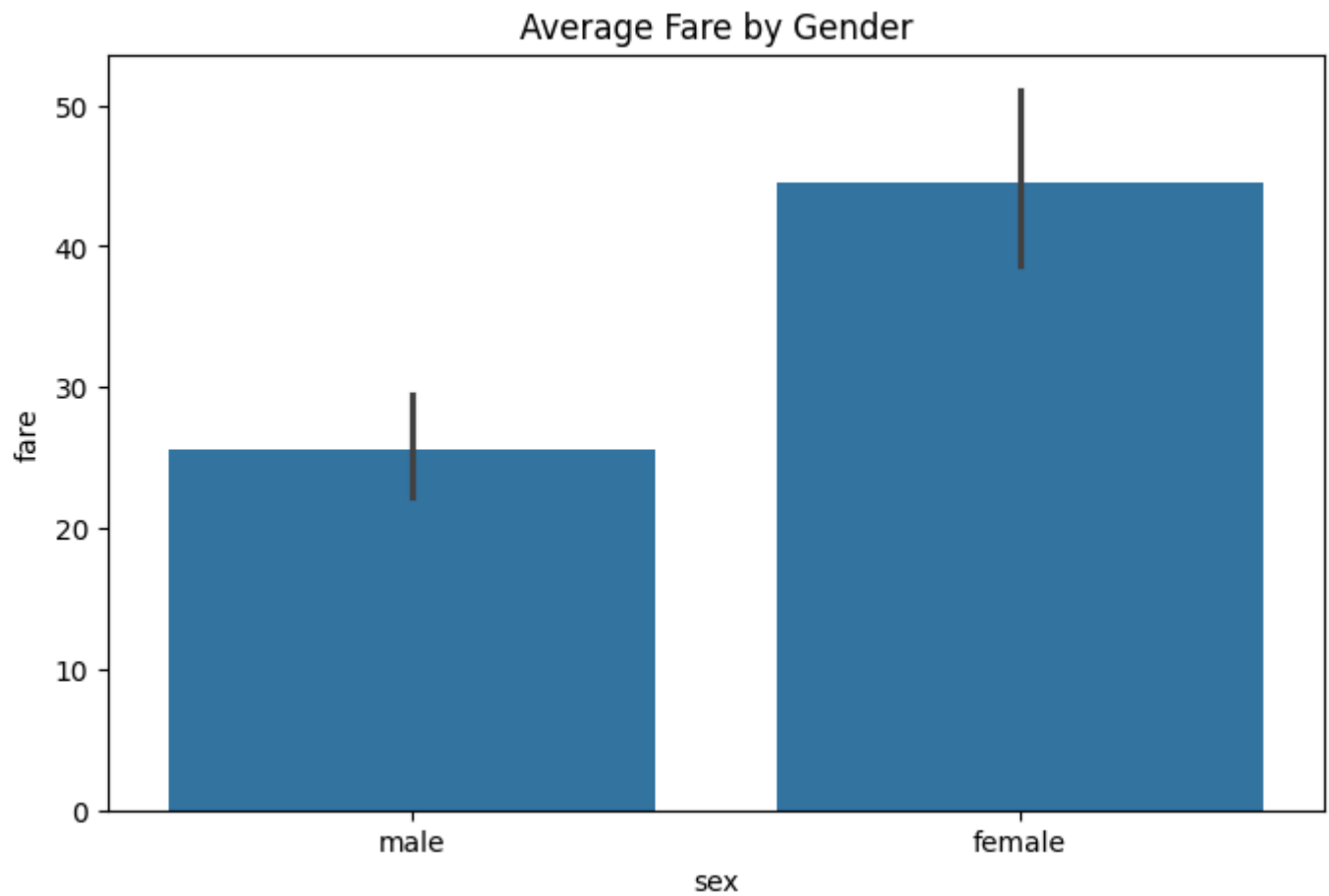
```
In [14]: sns.jointplot(x='age', y='fare', data=data, kind='scatter')
plt.show()
```



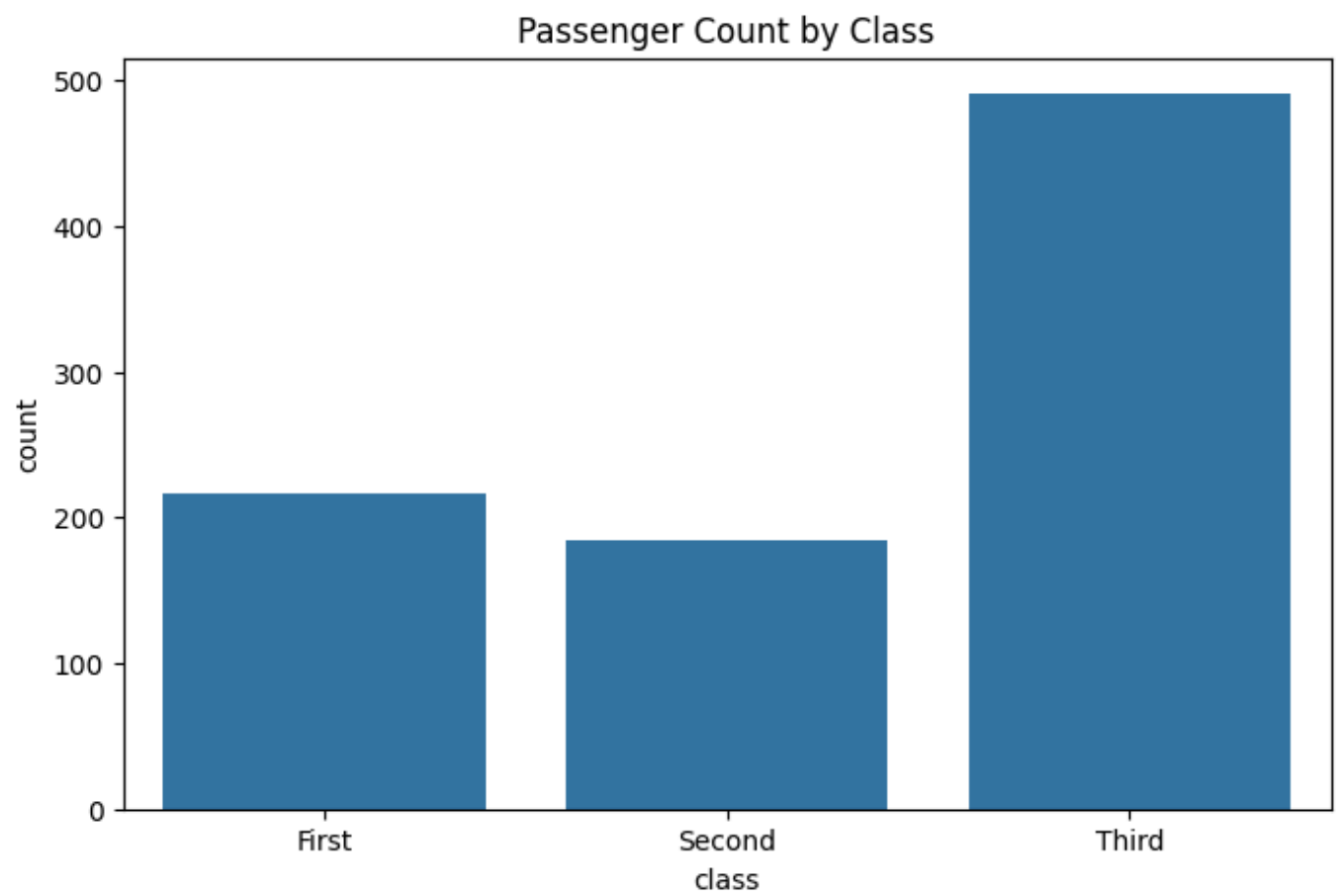
```
In [15]: plt.figure(figsize=(8,4))
sns.rugplot(data['fare'].dropna())
plt.title("Rug Plot of Fare")
plt.show()
```



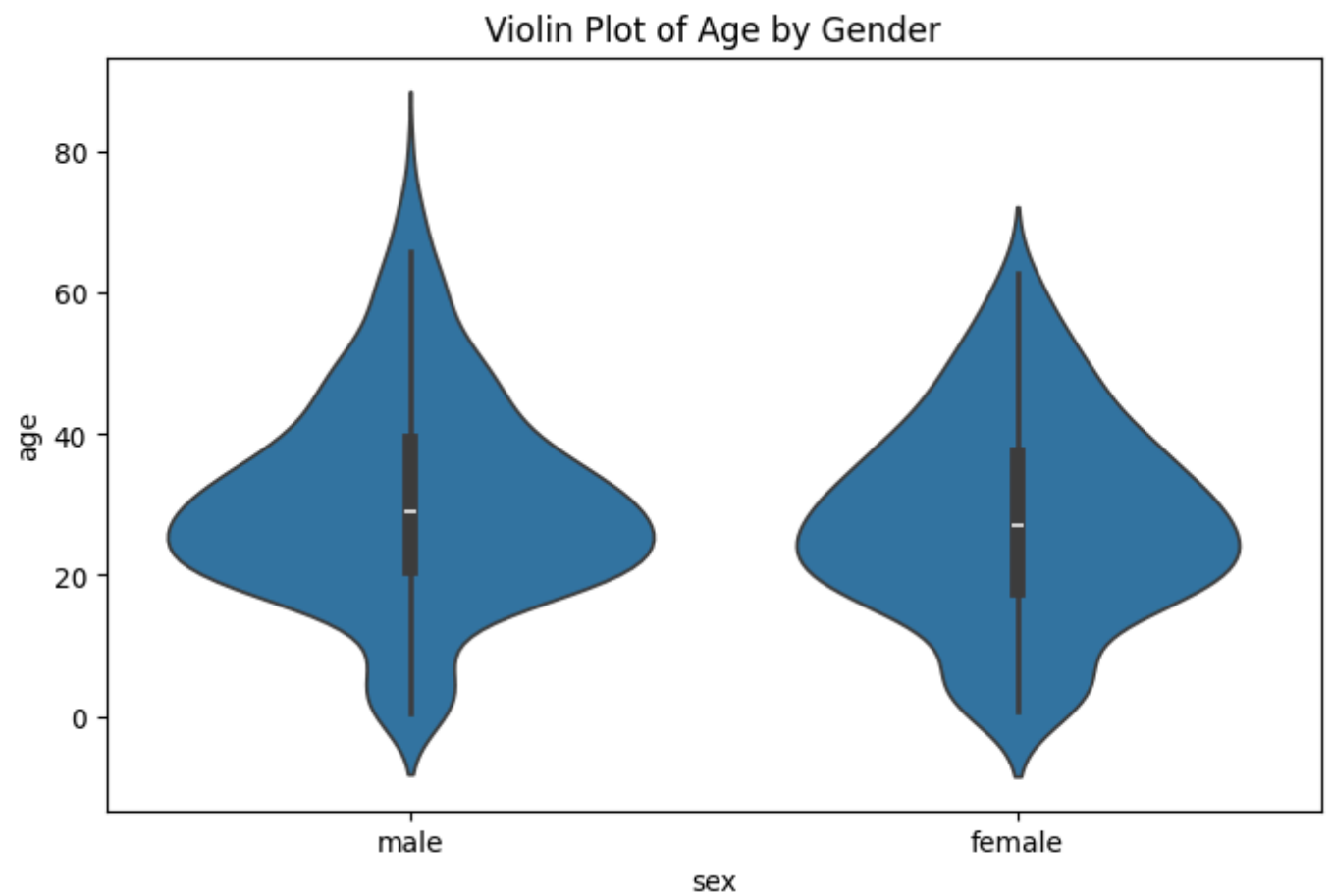
```
In [18]: plt.figure(figsize=(8,5))
sns.barplot(x='sex', y='fare', data=data)
plt.title("Average Fare by Gender")
plt.show()
```



```
In [19]: plt.figure(figsize=(8,5))
sns.countplot(x='class', data=data)
plt.title("Passenger Count by Class")
plt.show()
```

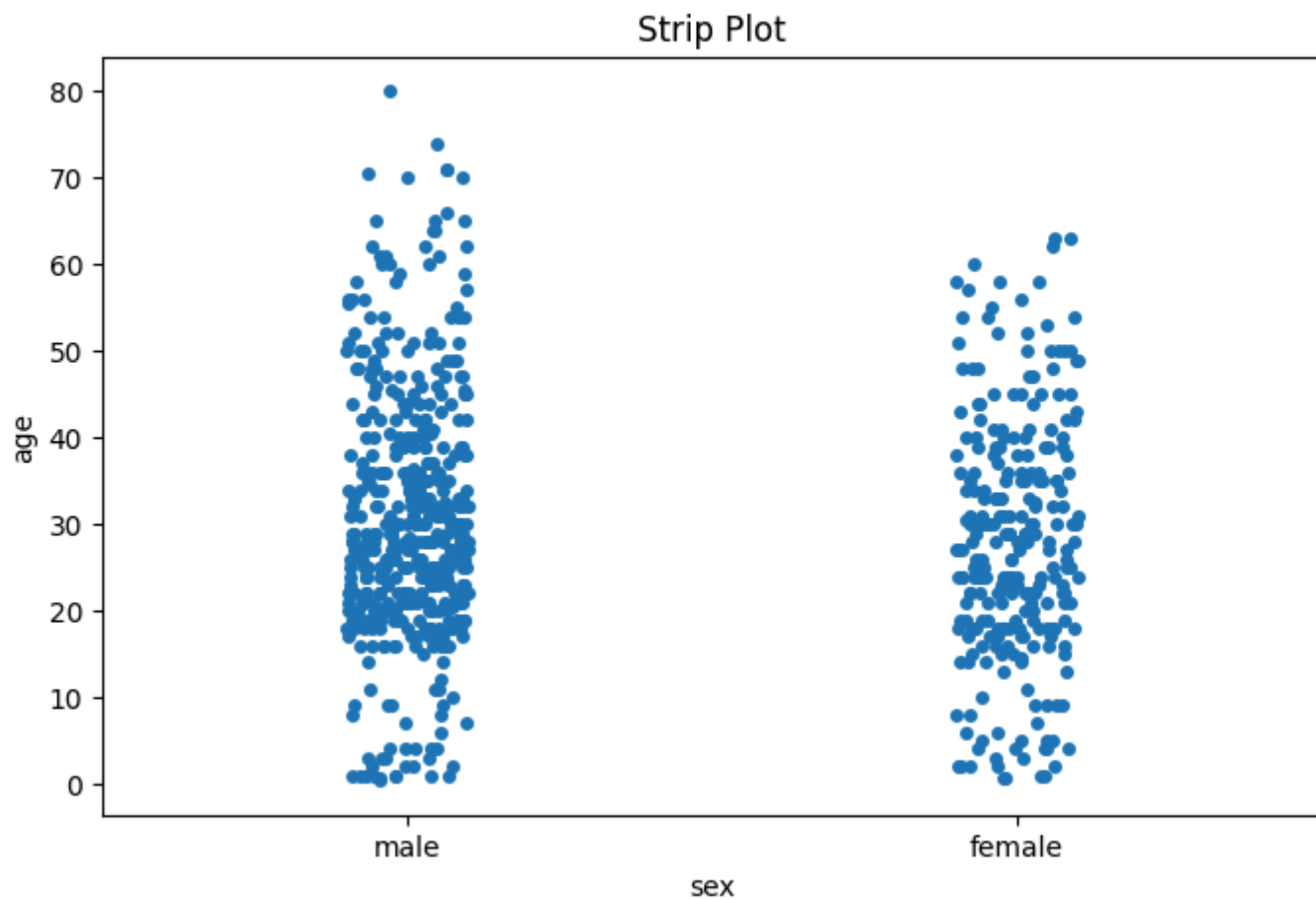


```
In [21]: plt.figure(figsize=(8,5))
sns.violinplot(x='sex', y='age', data=data)
plt.title("Violin Plot of Age by Gender")
plt.show()
```



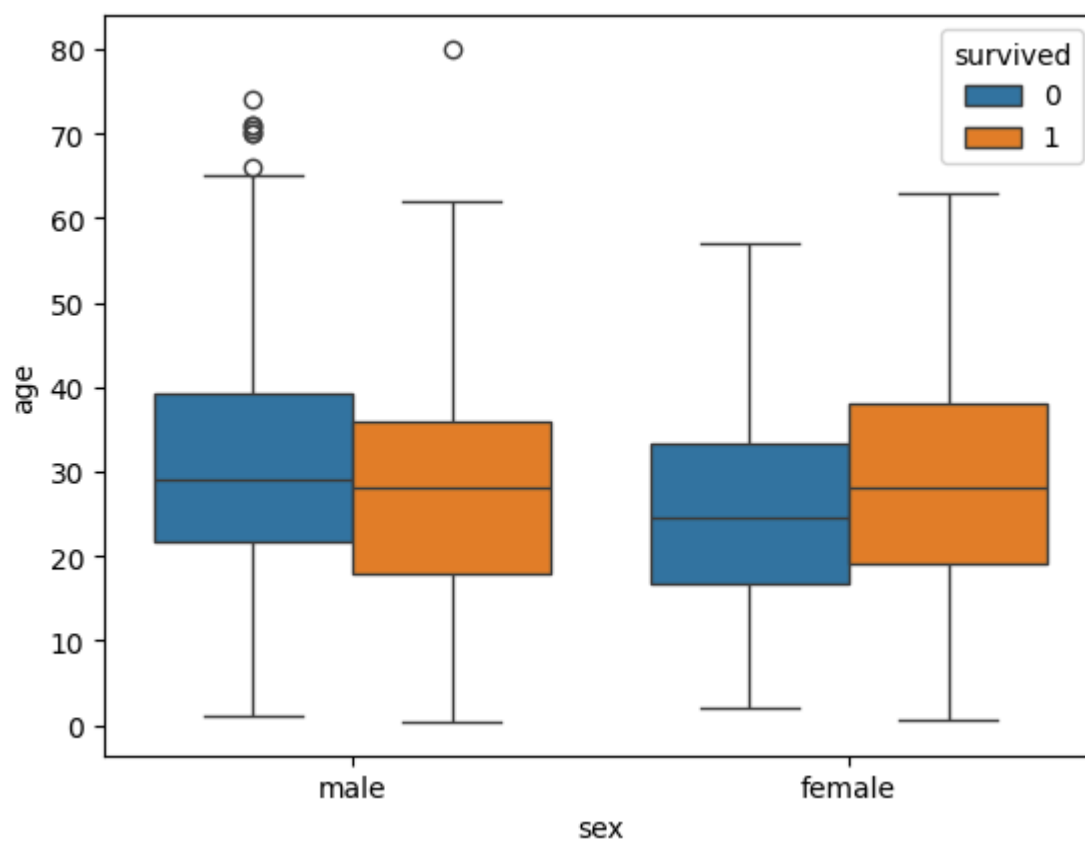
```
In [22]: plt.figure(figsize=(8,5))
sns.stripplot(x='sex', y='age', data=data)
```

```
plt.title("Strip Plot")  
plt.show()
```



```
In [30]: sns.boxplot(x='sex', y='age', hue='survived', data=data)
```

```
Out[30]: <Axes: xlabel='sex', ylabel='age'>
```



```
In [ ]:
```